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Genome-wide exploration of bZIP transcription factors and their contribution to alkali stress response in *Helianthus annuus*☆

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ABSTRACT

The basic leucine zipper (bZIP) gene family is one of the largest transcription factors family in plants. These are involved in various biological processes including the regulation of light signaling, seed maturation, flower development, and the response to both biotic and abiotic stresses. Despite the crucial role of this gene family, no comprehensive investigation has been carried out to characterize bZIPs in sunflower. Therefore, an *in silico* analysis along with a wetlab experimental approach was adopted for bZIPs in sunflower plants. During the study, 73 bZIPs were identified in sunflower. Phylogenetic analysis involving bZIPs from *A. thaliana, H. annuus, C. arabica*, L. *sativa,* and *V. vinifera* divided these members into 11 groups and testified using conserved motif and gene structure analysis. Protein-Protein interaction analysis identified five clusters of HabZIPs. Intron diversity ranged from 0 to 12. The *cis*-regulatory elements analysis predicted various stress-responsive elements. RNA-seq data analysis reveiled that clusters of bZIPs were involved in stress response in roots and leaves during early or late stages of plant growth. Similatly, qRT-PCR analysis identified that *HabZIP01, 04, 18,* and *48* were significantly upregulated in response to mild or svere salt stress. Conversely, *HabZIP08* was downregulated in response to both levels of salinity. Overall the study provides new insights into stress-responsive HabZIP genes in sunflower and provides a foundation for further research on their role in the plant's response to abiotic stress.

Introduction

Sunflower (*H. annuus*) is a plant of Asteraceae family, and it is considered one of the tallest annual plants, growing between 5 and 12 feet tall depending on the cultivar. North America is considered the primary center of origin for sunflowers. It is the third most important edible oil crop after soybean and cottonseed. This plant is also used as ornamental, animal feed, and raw material for the paper industry (Mantenese et al., 2006; Gonzáles Belo et al., 2017). Extreme climate conditions (abiotic stresses like salinity, heat stress, cold stress, drought, and biotic stresses like pathogen infections, insect infestation, nematode attacks, herbivory, or parasitic plants) can resist the oil percentage in seeds. As the temperature of the earth is increasing due to global warming and the salt contents in freshwater bodies (lakes, ponds, rivers, etc.) are also increasing (Lu et al., 2021). It is intensifying abiotic stress on plants and crops globally. There is a high need to develop new varieties of plants that are more resistant to abiotic stresses.

The bZIPs (Basic leucine zipper) is one of the largest transcription factor gene families in plants and is characterized by a highly conserved bZIP domain (Dröge-Laser et al., 2018). The members of this family play a major role in various biological processes like biotic and abiotic responses, flowering and plant development, seed development, and light signaling (Liu et al., 2023). These TFs work as regulators of many physiological processes like photomorphogenesis, and energy

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homeostasis. The bZIPs have been identified and characterized in several plants including Arabidopsis, tomato, rice, cucumber, maize, plum, Chinese pear, and castor bean (Yue et al., 2023). In Arabidopsis, the bZIP family comprises 75 members, which are further classified into 11 groups based on conserved motifs and sequence features (Dröge-Laser et al., 2018). The bZIP proteins have distinct nuclear localization signals and a conserved N-x7-R/K-x9 motif for DNA binding (Jain et al., 2018). They contain various heptad repeats of hydrophobic amino acids L-x6-L-x6-L. Members of the S group exhibit specific heterodimerization with members of the C group within the bZIP family. The largest group of bZIP transcription factors is Group S, which plays a vital role in carbohydrate metabolism and vascular systems. The group S of bZIP genes is also acknowledged in monocot and dicot species where they contribute to transcription activation. Cluster A is the second largest group of bZIP transcription factors and its members play a significant role in abscisic acid signaling and stress signaling in plants (Kobayashi et al., 2005). Group B is one of the most important groups in the family and acts as a vital regulator of the evolutionarily conserved endoplasmic reticulum (ER) stress response. This group is divided into two clusters based on their functional aspects. Cluster H is generally involved in the regulation of the expression of light-inducible genes by binding to G-boxes present in their promoters. It also plays a role in promoting photomorphogenesis, which is observed in light-grown mutant seedlings (Holm et al., 2002). Group X is one of the smallest groups and contains only five members. The members of Group I of bZIP genes play a significant role in vascular development and promote domain exchange with the GA3 gene from gibberellin-activated biosynthesis routes. Group E has some similarities to Group I as it possesses a zipper motif, but due to the absence of lysine in the 10th position, it is classified as a separate group (Nambara et al., 2013). Members of Group D are involved in defenses against pathogens and are involved in binding to the ethylene response element by interacting with AtEB. They also control floral organ numbers. Group G is involved in the regulation of light-responsive promoters associated with UV and blue light signal transduction. They may also play a role during seed development (Liu et al., 2019). Members of Cluster C comprise the extension of leucine zipper with up to nine heptad repeaters and the regulation of DNA-binding, which are preserved. They are also involved in the regulation of seed storage and protein manufacturing by interlinking with the PBF protein (Dröge-Laser et al., 2018).

Despite the significance of bZIPs for plant growth, development, and stress response, there is a clear lack of information on bZIPs in *H. annuus.* Therefore, current study was designed to identify and characterize bZIPs from *H. annuus.* For this purpose, a combination of bioinformatics and expression profiling techniques were used.

Materials and methods

Identification and sequence analysis of bZIPs in H. annuus

The bZIP sequences, encompassing genomic, transcriptomic, and proteomic data, were retrieved from multiple sources, including TAIR "https://www.arabidopsis.org" (Berardini et al., 2015), NCBI "https://www.ncbi.nlm.nih.gov/" (Coordinators, 2015), Phytozome "https://phytozome.jgi.doe.gov/pz/portal.html" (Goodstein et al., 2011), and Plant Transcription Factor Database "http://planttfdb.gao-lab.org/" (Dai et al., 2013). The blast searches were performed (using AtbZIPs as a query) for finding the homologs of bZIPs in *H. annuus* using the program BLASTP "(Protein BLAST: search protein databases using a protein query (nih.gov)" (Basic Altschul, 1990). All the retrieved protein sequences were analyzed using SMART database "(SMART: Main page (embl-heidelberg.de)" (Schultz et al., 2000) and UniProt "(https://www.uniprot.org; (2021)".

HabZIPs gene location on chromosome and gene ontology enrichment analysis

The NCBI gene database was utilized to identify the chromosomal positions of bZIPs in *H. annuus*. To examine gene duplication events, the TBtools "https://github.com/CJ-Chen/TBtools" sequence analysis toolkits were employed, considering various types of duplications such as tandem, segmental, and proximal, with a default criterion. The gene ontology enrichment analysis was performed by using an online server called ShinyGO "(http://bioinformatics.sdstate.edu/go/)" the locus tags of genes were retrieved from NCBI Gene database and are also listed in Gene characteristics Table 1.

Phylogenetic analysis of bZIPs transcription factors

The results of bZIPs protein blast against *H. annuus* and other plants like *coffee arabica, Lactuca sativa,* and *Vitis vinifera* were used to construct the phylogenetic tree with proteins of bZIPs. The Maximum Likelihood tree was constructed by using IQ-TREE "(http://iqtree.cibiv.univie.ac.at)" (Nguyen et al., 2015) based on 256 bZIP gene family sequences from *H. annuus, Coffee arabica, Lactuca sativa,* and *Vitis vinifera*. Furthermore, the tree was modified and visualized by using ITOL "(https://itol.embl.de/)" (Letunic and Bork, 2019).

Physiochemical characteristics of bZIP transcription factor proteins

The physiochemical characteristics of bZIPs were predicted using an online tool of the swiss institute of Bioinformatics called ProtParam "(https://web.expasy.org/protparam/)" (Garg et al., 2016). Motifs in the bZIP protein sequences were identified by using MEME (Multiple Em for Motif Elicitation) "(https://meme-suite.org/meme/tools/meme)" (Bailey et al., 2015) where site distribution was one sequence per occurrence and the number of motifs was selected to 10. Motif logos were also designed by MEME "(https://meme-suite.org/meme/tools/meme)". The subcellular localization of the bZIP transcription factor proteins was predicted using an online tool called ProtComp 9.0 "(http://www.softberry.com/berry.phtml?topic=protcomppl&group=programs&subgroup=proloc)" (Hategan and Tabus, 2007). bZIP protein domains were predicted using Interpro "(https://www.ebi.ac.uk/interpro)" (Mulder and Apweiler, 2007) and were aligned using DNAMAN software. "(https://www.lynnon.com/)" (Wang, 2015).

Gene sequence analysis of bZIP transcription factors in H. annuus

Exon and intron structures of bZIP transcription factors genes in *H. annuus* were predicated using GSDS (Gene Structure Display Server v2.0) "(http://gsds.gao-lab.org/)" (Hu et al., 2015).

HabZIPs domains and protein-protein interaction analysis

For determination of presence of conserved domains in bZIP transcription factors in *H. annuus* further analysis phases were taken for visualizing the domains in HabZIPs. The NCBI Conserved Domain Database (CDD) "(https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi)" (Marchler-Bauer et al., 2015) was used to determine the domains present in HabZIP proteins. After retrieving the results, the domains were visualized using TBtools (Chen et al., 2020). PPI network of HabZIP proteins was constructed using the STRING database "(https://string-db.org/)" (Mering et al., 2003). After giving input of sequences to the database, the protein network with other proteins was constructed. After that, the k-means clustering algorithm was applied to the protein-protein interaction network (Ahmad et al., 2020).

Identification of cis-acting regulatory elements of habZIPs genes

For analyzing the *cis*-regulatory elements in *H. annuus,* a 2 kb

Table 1
Genomic and physiochemical details of HabZIPs.

Gene id	Protein Name	Protein ID	No of Amino acids	Molecular weight	Theoretical PI	Exon	Gravy
LOC110936230	HabZIP01	XP_022034264.1	160	17,951.93	5.91	2	−0.884
LOC110895426	HabZIP02	XP_021998431.1	2899	257,645.97	6.45	1	0.557
LOC110897054	HabZIP03	XP_021999669.1	183	21,508.85	5.62	1	−0.972
LOC110909336	HabZIP04	XP_022010055.1	2437	217,008.14	6.03	1	0.522
LOC110895443	HabZIP05	XP_021998452.1	183	21,331.21	10.14	1	−0.868
LOC110874584	HabZIP06	XP_021978839.1	174	20,219.57	9.77	1	−0.994
LOC110867841	HabZIP07	XP_021972584.1	190	22,215.86	9.79	1	−0.93
LOC110909570	HabZIP08	XP_022010148.1	164	19,107.39	6.38	1	−0.765
LOC110930191	HabZIP09	XP_022029119.1	5563	481,413.83	5.53	6	0.587
LOC110912196	HabZIP10	XP_022012561.1	7179	617,911.97	5.59	6	0.604
LOC110885891	HabZIP11	XP_021989300.1	175	19,245.39	6.28	1	−0.507
LOC110865426	HabZIP12	XP_035830631.1	5359	459,264.15	6.33	5	0.531
LOC110889069	HabZIP13	XP_021992259.1	222	25,006.32	9.37	6	−0.815
LOC110912248	HabZIP14	XP_022012616.1	4012	349,280.43	7.07	3	0.604
LOC110914686	HabZIP15	XP_022015158.1	4470	382,908.35	6.53	5	0.483
LOC110930079	HabZIP16	XP_022029000.1	382	40,875.69	5.98	13	−1.047
LOC110897391	HabZIP17	XP_021999833.1	6624	572,849.2	6.17	2	0.531
LOC110878933	HabZIP18	XP_021983020.1	7000	615,056.72	6.19	4	0.457
LOC110920959	HabZIP19	XP_022020900.1	707	76,357.82	6.05	7	−0.445
LOC110908345	HabZIP20	XP_022008966.1	335	37,343.03	5.96	10	−0.888
LOC110898101	HabZIP21	XP_022000526.1	273	30,181.96	6.31	14	−0.662
LOC110889076	HabZIP22	XP_021992267.1	7309	629,708.25	6.82	10	0.604
LOC110898088	HabZIP23	XP_022000511.1	273	30,059.57	5.81	3	−0.781
LOC110920619	HabZIP24	XP_022020485.1	4365	378,707.3	5.85	3	0.479
LOC110899302	HabZIP25	XP_022001884.1	6287	544,316.11	5.35	6	0.62
LOC110865794	HabZIP26	XP_021970810.1	328	36,462.02	8.63	10	−0.53
LOC110939945	HabZIP27	XP_035833450.1	3964	347,475.31	7.2	3	0.606
LOC110928563	HabZIP28	XP_022027269.1	6251	544,955.89	6.1	4	0.555
LOC110922794	HabZIP29	XP_022022682.1	584	63,658.61	7	5	−0.977
LOC110911549	HabZIP30	XP_022011868.1	456	50,533.01	6.29	5	−0.695
LOC110921965	HabZIP31	XP_022021980.1	4878	422,331.26	6.6	4	0.507
LOC110911549	HabZIP32	XP_022011869.1	447	49,786.61	7.13	5	−0.606
LOC110911549	HabZIP33	XP_022011870.1	6545	564,885.64	6.73	5	0.459
LOC110929288	HabZIP34	XP_022028109.1	269	30,209.34	5.64	4	−0.861
LOC110889853	HabZIP35	XP_021993119.1	5573	481,913.51	6.57	6	0.537
LOC110936130	HabZIP36	XP_022034155.1	422	45,165.57	9.69	5	−0.582
LOC110879426	HabZIP37	XP_021983576.1	5764	489,754.45	6.36	5	0.469
LOC110889853	HabZIP38	XP_021993117.1	5579	482,543.22	6.64	6	0.537
LOC110889315	HabZIP39	XP_021992515.1	444	47,959.69	9.13	5	−0.713
LOC110890308	HabZIP40	XP_021993593.1	5966	511,650.26	6.43	4	0.611
LOC110929141	HabZIP41	XP_022027954.1	8821	758,772.44	5.93	12	0.59
LOC110911291	HabZIP42	XP_022011585.1	187	21,892.35	7.86	1	−0.943
LOC110879036	HabZIP43	XP_021983133.1	2190	196,221.29	5.3	1	0.59
LOC110915267	HabZIP44	XP_022015627.1	152	17,598.65	6.42	1	−0.971
LOC110895842	HabZIP45	XP_021998898.1	326	36,480.08	8.63	9	−0.538
LOC110923052	HabZIP46	XP_022022931.1	6712	577,094.44	6.68	11	0.544
LOC110912966	HabZIP47	XP_022013506.1	7313	622,664.81	5.77	9	0.606
LOC110914945	HabZIP48	XP_022015238.1	2237	202,518.58	5.48	1	0.498
LOC110928563	HabZIP49	XP_022027268.1	750	81,215.81	5.67	4	−0.456
LOC110918354	HabZIP50	XP_022018389.1	6488	558,424.12	6.68	9	0.641
LOC110942245	HabZIP51	XP_022039687.1	6678	576,386.89	6.58	4	0.61
LOC110869038	HabZIP52	XP_021974031.1	6476	554,428.02	6.12	4	0.588
LOC110864612	HabZIP53	XP_021969419.1	148	16,695.84	6.12	1	−0.665
LOC110909213	HabZIP54	XP_022009937.1	7277	627,966.82	6.48	12	0.561
LOC110909213	HabZIP55	XP_022009935.1	379	40,738.31	6.48	12	−0.88
LOC110923683	HabZIP56	XP_022023437.1	380	40,809.39	6.48	4	−0.873
LOC110878290	HabZIP57	XP_021982260.1	159	17,573.61	9.73	8	−1.066
LOC110921952	HabZIP58	XP_022021970.1	359	40,422	5.96	1	−0.416
LOC110878582	HabZIP59	XP_021982615.1	7478	654,329.26	6.44	5	0.525
LOC110907992	HabZIP60	XP_022008590.1	169	19,766.98	5.76	3	−0.851
LOC110929250	HabZIP61	XP_022028081.1	262	30,059.98	4.88	4	−0.535
LOC110917891	HabZIP62	XP_022018030.1	6085	526,254.63	6.92	7	0.596
LOC110878652	HabZIP63	XP_021982684.1	313	34,245.63	5.11	6	−0.78
LOC110898577	HabZIP64	XP_022001067.1	2626	227,024.51	7.41	3	0.414
LOC110885377	HabZIP65	XP_021988761.1	522	58,166.12	6.39	11	−0.624
LOC110938001	HabZIP66	XP_022036159.1	298	33,175.57	9.72	4	−0.788
LOC110873324	HabZIP67	XP_021977945.1	382	41,733.82	9.26	6	−0.73
LOC110899342	HabZIP68	XP_022001925.1	372	40,127.03	5.64	13	−1.025
LOC110872654	HabZIP69	XP_021977191.1	333	36,436.1	5.84	4	−0.759
LOC110922332	HabZIP70	XP_022022344.1	210	24,464.09	5.71	1	−0.894
LOC110921965	HabZIP71	KAF5754620.1	4923	427,584.39	6.68	4	0.51
LOC110922844	HabZIP72	XP_022022734.1	280	31,957.38	5.63	5	−0.946
LOC110883864	HabZIP74	XP_021987208.1	296	32,825.7	6.73	4	−0.765

promoter region was taken manually from the NCBI nucleotide database for each of the HabZIP genes "(Home - Gene - NCBI (nih.gov)" (Brown et al., 2015). After retrieving the 2000 bp of promoter region the PlantCare database "(http://bioinformatics.psb.ugent.be/webtools/plantcare/html/)" (Lescot et al., 2002), a database of plant *cis*--acting regulatory elements and a portal to tools for *in silico* analysis of promoter sequences were used to identify the *cis*-acting regulatory elements in HabZIPs.

Expression profiling using RNA-seq data

The raw transcriptomic data for the alkali stress (SRS7799829) was retrieved from the NCBI SRA website "(https://www.ncbi.nlm.nih.gov/sra/SRX9595538)" (Leinonen et al., 2010). The reads were then mapped to the reference genome of *H. annuus* using the Galaxy Bowtie2 with default parameters "(https://usegalaxy.org/)" (de Almeida et al., 2016). The expression levels of genes were estimated using Cufflinks, a tool that calculates FPKM values from RNA-Seq data. The resulting FPKM values were extracted from the output files and used to identify differentially expressed genes between the control and alkali stress conditions (Waqas et al., 2019). The resulting gene list was incorporated into an analysis for building the expression profiles using the TBtools software.

RNA extraction and RT-QPCR of habZIPs

RNA was isolated from *H. annuus* leaves using Trizol reagent and quantified using a Nanodrop 2000 (Thermo Fischer Scientific, Waltham, MA, USA). One µg of RNA was used to synthesize cDNA using the First-Strand Synthesis kit. The cDNA was stored at −20 °C for future use. Performed qRT-PCR using an iTaq Universal SYBR Green Super-Mix (Bio-Rad Labs, Hercules, CA, USA) and a qRT PCR detection system (CFX96 Touch Real-Time PCR Detection System). The NCBI Primer-BLAST algorithm "(https://www.ncbi.nlm.nih.gov/tools/primer-blast/ (accessed on 13 July 2022))" was used to confirm gene-specific primers designed using Oligo Calculator "(http://mcb.berkeley.edu/labs/krantz/tools/oligocalc.html (accessed on 13 July 2022))". Each treatment was performed in triplicates and the expression of each gene was measured three times in each sample with *HaGAPDH* (*Glyceraldehyde-3-phosphate dehydrogenase*) serving as a housekeeping gene.

Results

Identification and phylogenetic analysis of bZIP transcription factors in H. annuus

A total of 73 members of bZIP transcription factors were identified in *H. annuus*. For confirming the occurrence of a signature domain in these sequences, an online tool called SMART "(http://smart.embl-heidelberg.de)" (Schultz et al., 2000) was used. Information related to bZIPs in both species is present in Supplementary Table S1. The bZIPs in *H. annuus* were named according to their homologs in *A.thaliana*.

To illustrate the evolutionary relationship among the bZIPs from *H. annuus* and other plants like *C. arabica,* L. *sativa, and V. vinifera,* bZIP protein sequences from all five species were aligned using the MEGA 7.0.21 "(https://www.megasoftware.net)" (Kumar et al., 2016) (Fig. 1). The alignment results were used to generate the Maximum likelihood tree by using IQtree (Nguyen et al., 2015).

All the bZIPs from *A. thaliana, H. annuus, C. arabica,* L. *sativa,* and *V. vinifera.* were allocated to eleven subfamilies (Fig. 1) with their orthologs from *A. thaliana* and these subfamilies are in good accordance to *A, thaliana* based classification that has already been reported in previous studies (Llorca et al., 2014; Nazri et al., 2017; Kumar et al., 2018). Group X represents the uncharacterized members of bZIPs homologs. Group S represents the largest group of HabZIPs with a total of 81 members from *A. thaliana, H. annuus, C. arabica,* L. *sativa,* and

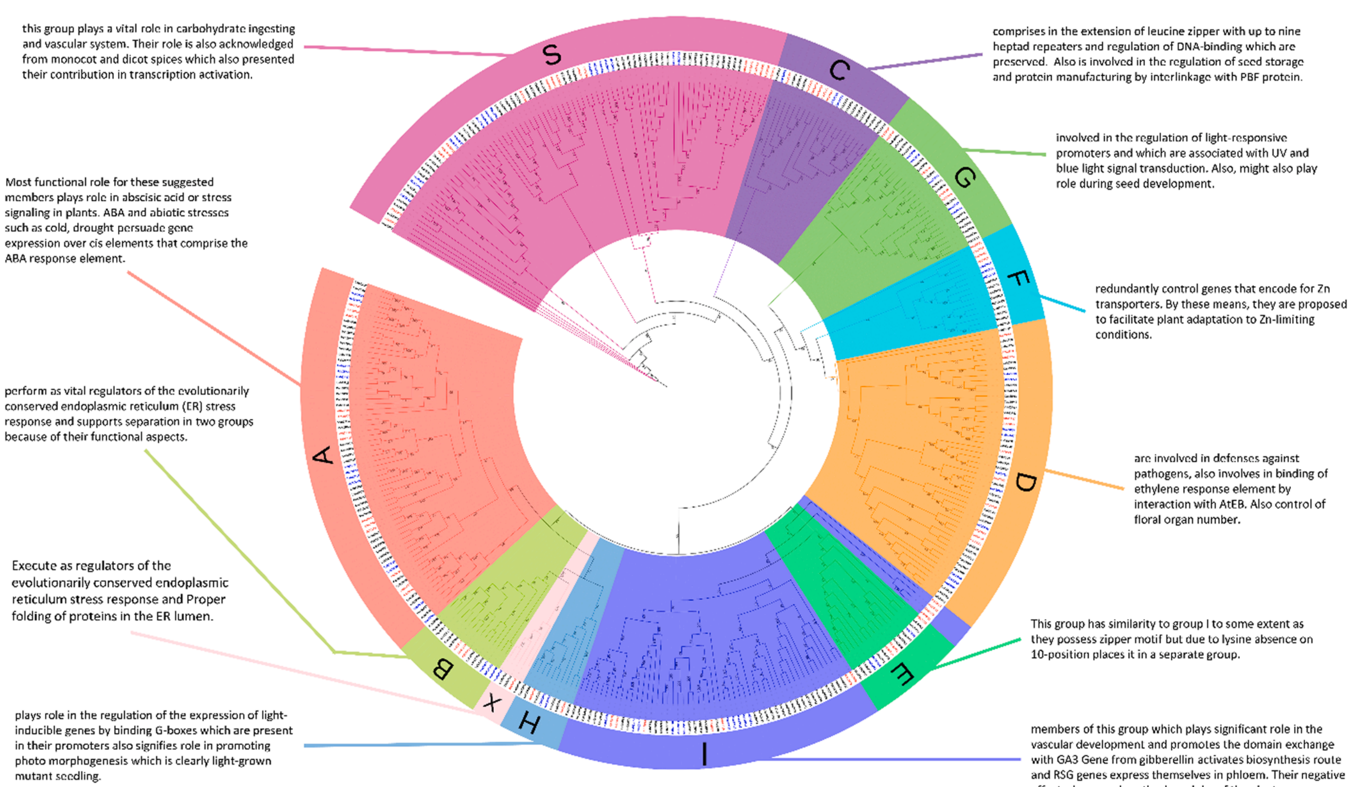

Fig. 1. Phylogenetic tree of bZIPs from *A. thaliana, H. annuus, C. arabica, L. sativa, and V. vinifera*. Using IQ-TREE webserver the maximum likelihood tree was generated and a total of 256 protein sequences were used in phylogenetic analysis. The blue color in tree represents *H. annuus* and the red color is representing the reference sequences of bZIPs which is of *A. thaliana*.

V. vinifera. The sequences with blue labels in phylogenetic tree represent the sequences of *H. annuus* and the red-colored labels represent the sequences of *A. thaliana.* Group H represents the smallest number of clades in phylogenetic tree and contains only ten members of all the corresponding species. All the clades contain at least one member of all the corresponding species sequences (Fan et al., 2019).

HabZIPs protein characteristics analysis

The basic physiochemical properties of 73 HabZIPs proteins were studied (Table 1) the results demonstrated that amino acid numbers ranged from 152 to 8821. The HabZIP41 protein contains the largest number of amino acids that ranged to 8821 and the smallest number of amino acids were present in the HabZIP44 that ranged to 152 amino acids only. 33 Proteins were present in the extracellular location the proteins present outside the cell membrane are usually called extracellular proteins. All the other proteins were located in nucleus. Theoretical isoelectric points ranged from 5.11 for HabZIP63 to 10.14 for HabZIP05. All the HabZIP proteins were hydrophilic, with the GRAVY (Grand Average of hydropathicity) below 0 (Azeem et al., 2021a). The Exon was also included in the table that were identified from NCBI Gene page and the highest number of exons were 14 of *HabZIP21*. Conserved domains in HabZIPs were identified using InterPro, and their characteristics were determined based on previous studies (Higgins and Novak, 1997).

HabZIPs gene distribution on chromosomes and gene ontology enrichment analysis

Based on the genomic information obtained from NCBI-Gene, positions of all genes were visualized on chromosomes using phenogram by Ritchie lab "(http://visualization.ritchielab.org/phenograms/plot)" (Wolfe et al., 2013). The basic input files were divided into two notepad files and basic visualization of gene location on chromosome was performed using web-based tool phenogram. After obtaining all the basic visualizations, the image was further modified in Microsoft Paint software available on Windows OS. The final image, after further modifications, shows that only two members of HaBZIPs genes are located on chromosomes one and eight. While the highest number of genes were present on chromosomes 9 and 17. Each of the other chromosomes contains roughly 8 gene members of HabZIPs, and the colored dots show the 11 groups of HabZIPs. All the groups contain a specific color by which it can easily identify the gene group position on chromosome (Fig. 2A).

HabZIP genes were involved in three major gene enrichment processes i.e., Biological processes, Molecular Functions, and Cellular Components (Fig. 2B). A major number of genes were involved in biological functions such as regulation of photomorphogenesis, Response to far-red light, Regulation of response to red or far-red light, Red or far-red light signaling pathway, Endoplasmic reticulum unfolded protein response, Positive regulation of transcription, positive regulation of RNA biosynthetic proc, Positive regulation of nucleic acid-templated transcription, positive regulation of nucleobase-containing compound metabolic process functions. HabZIP genes are involved in cellular components as well screening these functions, such as the perinuclear region of cytoplasm, Endoplasmic reticulum membrane, Nuclear outer membrane-endoplasmic reticulum membrane network, Endoplasmic reticulum sub-compartment, Organelle sub-compartment functions. Genes involving molecular processes such as DNA-binding transcription factor activity, Transcription regulator activity, Transcription *cis*-regulatory region binding, Transcription regulatory region nucleic acid binding, Sequence-specific DNA binding, Sequence-specific double-stranded DNA binding, Double-stranded DNA binding, DNA-binding transcription factor activity and RNA polymerase II-specific functions.

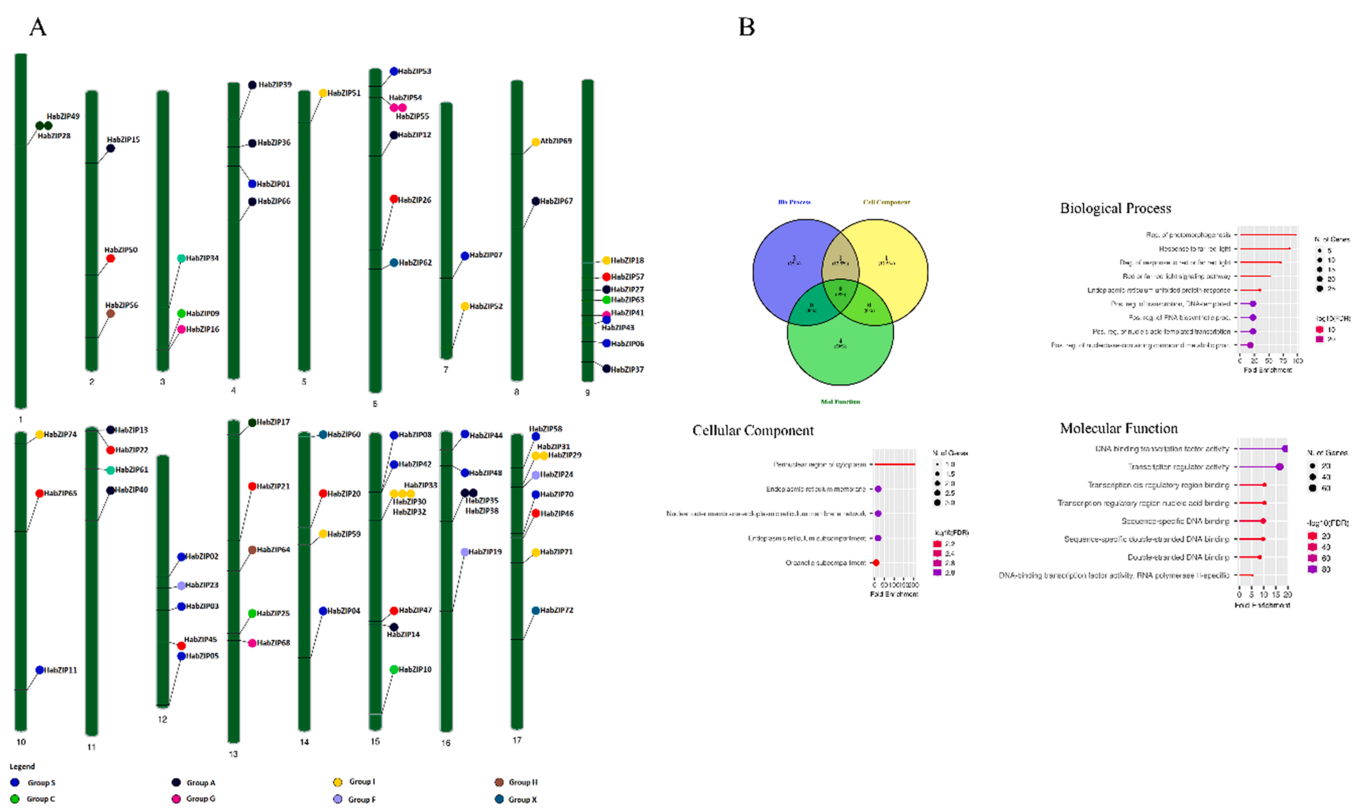

Fig. 2. (A) Distribution of HabZIP genes on *H. annuus* on 17 chromosomes. The 73 HabZIP genes were mapped onto 17 chromosomes and the genes are divided into 11 groups. (B) Gene ontology enrichment analysis of HabZIP genes demonstrating how many functions genes are performing.

HabZIPs domains and protein-protein interaction analysis

The conserved domains present in all the bZIPs of *H. annuus* were predicted using interpro and NCBI CDD present in the (Table 1) and then the further analysis phases were taken and the visualization of domains was done through using TBtools. All the proteins in *HabZIPs* contain the conserved domains i.e., (bZIP_Plant_Bzip46, BZIP_HBP1b_like, bZIP_1, bZIP_plant_GBF1, bZIP_HY5_like) are the domains conserved in all the HabZIPs. However, additional domains were also identified in the bZIP transcription factors of *H. annuus* through visualization using TBtools. As discussed the most conserved domains of HabZIPs. The least conserved domain present is ZIP_TSC22D-like Superfamily which is present only in HabZIP16 which shows this protein is distantly functional in any other protein families (Fig. 3A).

PPI Network of HabZIPs was constructed using String database and k-means clustering algorithm was also applied to proteins. That divided the proteins into 5 clusters. All The bZIPs proteins were interacting with each other in each cluster (Fig. 3B). Meanwhile, there were two proteins of HabZIPs that were not interacting which were bZIP3 and bZIP23 in cluster 5 which is represented by cyan color.

Gene structure and conserved motifs analysis of H. annuus bZIPs

For an enhanced understanding of the structural features of HabZIPs genes, we analyzed the intron and exon structures of *H. annuus* bZIPs transcription factors using Gene Structure Display Server 2.0. The results of analysis demonstrate that the number of introns varies from 0 to 12. The results shown in (Fig 4B) clearly explain there is a certain group of intronless genes. These genes belong to group S. All the remaining groups contain almost 1- 8 introns regions. Meanwhile, most of the introns exist in group G where the number of introns ranges from 1 to 12. Three Genes of group I contains upstream and downstream region represented by red color in the (Fig. 4) legend. For gaining insight into the divergence of HabZIPs proteins, conserved motifs were predicted using the MEME Program (https://meme-suite.org/meme/), and the ungapped motifs were identified where the site distribution was selected to one occurrence per sequence, and the number of motifs was selected to 10. In all the groups of bZIPs of *H. annuus* motif 1,2 and 3 was conserved in all the bZIPs of *H. annuus* as represented in (Fig. 4C). All the renaming motifs were present in some of the proteins as can be seen in the (Fig. 4C). The least conserved motif present is motif 10 present in some of the proteins of bZIPs of *H. annuus* (Lewi et al., 2006). Which clearly explains the functional integrity of motif. Mainly a motif is a conserved sequence in proteins that explains the distinct function. The legend clearly explains the motif number, their symbol on the figure, and Motif conserved sequence (Adeleke and Babalola, 2020).

Cis-acting regulatory elements of Habzip genes

Cis-acting regulatory elements are molecular switches that are involved in the transcriptional regulation of genes numerous biological phenomena like development processes, hormones, and stress response (Hernandez-Garcia and Finer, 2014). The *cis*-acting regulatory elements include enhancers, silencers, or insulators (Rombauts et al., 1999). The promoters of all the HabZIPs contain TGACG-motif (Motif Binding Basic Leucine Zipper Transcription Factors) (Kumar et al., 2012), Sp1 (involved in light responsiveness), and G-box (The ubiquitous G-box or CACGTG) (Menkens et al., 1995). Proteins known as G-box factors (GBFs) bind to G-boxes in a context-specific manner, mediating a wide variety of gene expression patterns. Moreover, TATA-box was also detected in all HabZIPs. Additionally, the abiotic stress response elements like LTR, ABRE, HD-ZIP, and MBS were identified in variable numbers (Fig. 5). The ABA-responsive element (ABRE; PyACGTG/TC) as discussed earlier in the introduction is a *cis*-acting element involved in the abscisic acid responsiveness. Multiple (ABRE/ABA) elements are required for the ABA-responsive transcription to occur. The ABRE-Binding Protein/Factor (ABRE/ABF) family belongs to group A of the basic leucine zipper (bZIP) transcription factor family. Therefore, it is inferred that members of bZIPs are potentially involved ABA dependent stress response (Jin et al., 2010).

Gene expression pattern of bZIP transcription factors in leaves of H. annuus against alkali stress

Alkali stress is a complex type of plant stress that can cause various negative effects on plants, such as chemical destruction and osmotic

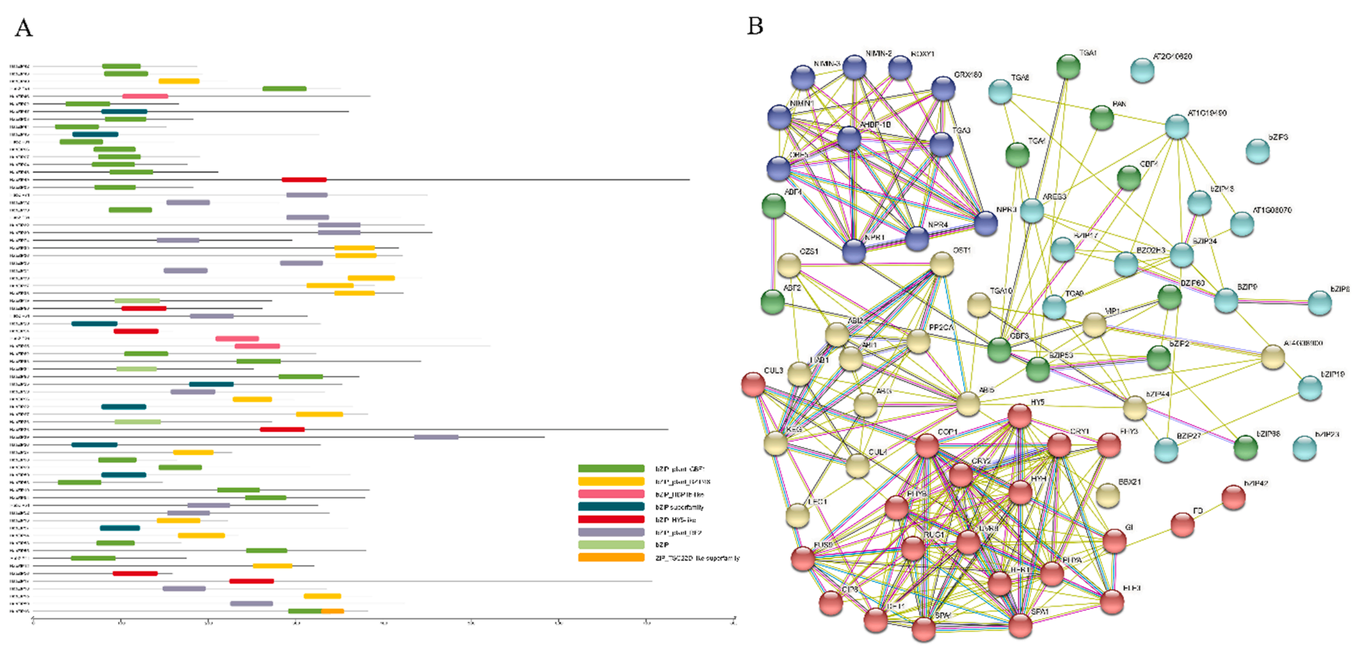

Fig. 3. (A) HabZIPs proteins domains are being displayed and are retrieved from NCBI CDD bZIP-plant-GBF1 is the most conserved domain in the proteins. (B) The protein-protein interaction network which is developed by String database is shown in B part of figure. bZIP3 and bZIP23 are the only proteins that are not interacting in the network.

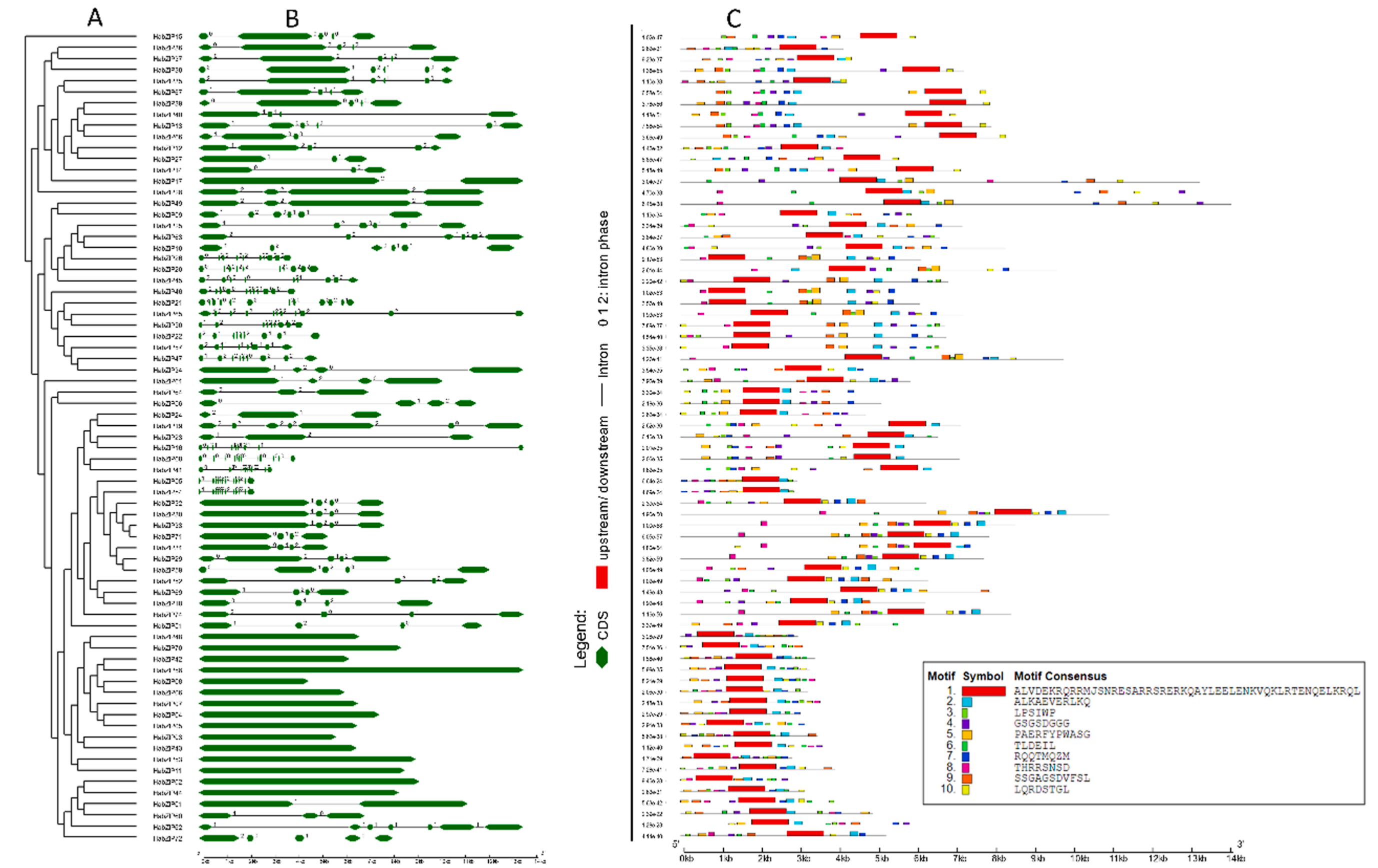

Fig. 4. (A) The phylogenetic tree of sequences is also displayed so their clustering is visualized. (B) Exon and Intron Structures of bZIPs were characterized using GSDS 2.0 webserver. Exons are shown are green rectangles with sharp end edges and introns with black lines. Intron stages are also represented by 0,1, and 2. (C) Motifs are also represented in figure the ten most conserved motifs are represented on all proteins and their sequences are also shown in figure below.

stress. Soil alkalization can also cause severe problems in some areas (Yang et al., 2009). Although sunflower is an important oil seed crop, and recent studies showed that it has strong alkali tolerance, however, it is not widely understood. In this study, we used next-generation sequencing (NGS) data to analyze gene expression in *H. annuus* roots under alkali stress (NaCl:NaOH = 9:1; pH 7). A total of six runs were observed under bio project PRJNA681139. A total of 11.855 gigabytes of experimental data was uploaded to Sequence Read Archive (SRA) with the accession number SRP104063. The experimental data were used to evaluate the FPKM values by using the Galaxy server-based tool and produced almost 30.15 gigabytes of data with more than 15,211, 756 spots. We calculated the FPKM values for all of the HabZIP genes and used the TBtools to create a heatmap (Fig. 6A) (Waqas et al., 2020). Overall, bZIPs respond in clusters to stress in early days (25 days), mid days (50 days) and late (75 days). Interestingly, no bZIP was upregulated from beginning to end of experiment. However, only four bZIPs including *HabZIP08* were significantly downregulated during whole experiment.

Gene expression pattern of bZIP transcription factors in roots of H. annuus against alkali stress

The runs were identified in the same experiments and the data was collected from bio project PRJNA681139. The ongoing study comprises a total of six runs of alkali stress of which three of them are control and the remaining three are stress runs (Ali et al., 2018; Azeem et al., 2021b). Expression profiles of bZIPs were divided into groups of early or late response clusters (Fig. 6B). Almost 20 bZIPs including *HabZIP23,30* and *60* were upregulated after 25 days, almost similr number of HabZIPs were upregulated after 50 days or 75 days. Moreover, these clusters were generally differentially regulated for a specific period only.

Expression analysis of HabZIPs genes

Using real-time amplification (qRT-PCR) the transcript abundance of 9 of the HabZIP genes from *H. annuus* leaves was calculated. The qPCR-based quantification of *HabZIP48, HabZIP18, HabZIP60, HabZIP08, HabZIP01, HabZIP04, HabZIP05, HabZIP06,* and *HabZIP42* was performed. Differential regulation of all HabZIPs was observed during NaCl stress. *HabZIP01, 04, 18,* and *48* were differentially upregulated in response to both mild and severe stress of 50 mM and 100 mM, respectively (Fig. 7). On the other hand, *HabZIP05, 06,* and *42* were upregulated only in response to 100 mM stress. *HabZIP60* was upregulated in response to mild stress and downregulated in response to 100 mM stress. Interestingly, *HabZIP08* was downregulated in response to both stresses (Fig. 7).

Discussion

Sunflower is an important oil extraction source, used as pigmentation, for various medical purposes, and as decorative plants. Compared to other plants, it grows over high latitudes and in high temperature and drought conditions which has made it unique. Due to its distinct features, it has great importance all over the world. Severe salt or drought conditions can cause depletion in oil and seed production (Ramadan et al., 2019; Azeem et al., 2020). Study revealed a lack of available data concerning salt stress and important transcription factor families in sunflower. The bZIP transcription factor gene family is involved in various biological processes like synchronization of plants, its development, light indication as well as biotic and abiotic stresses (Liu et al.,

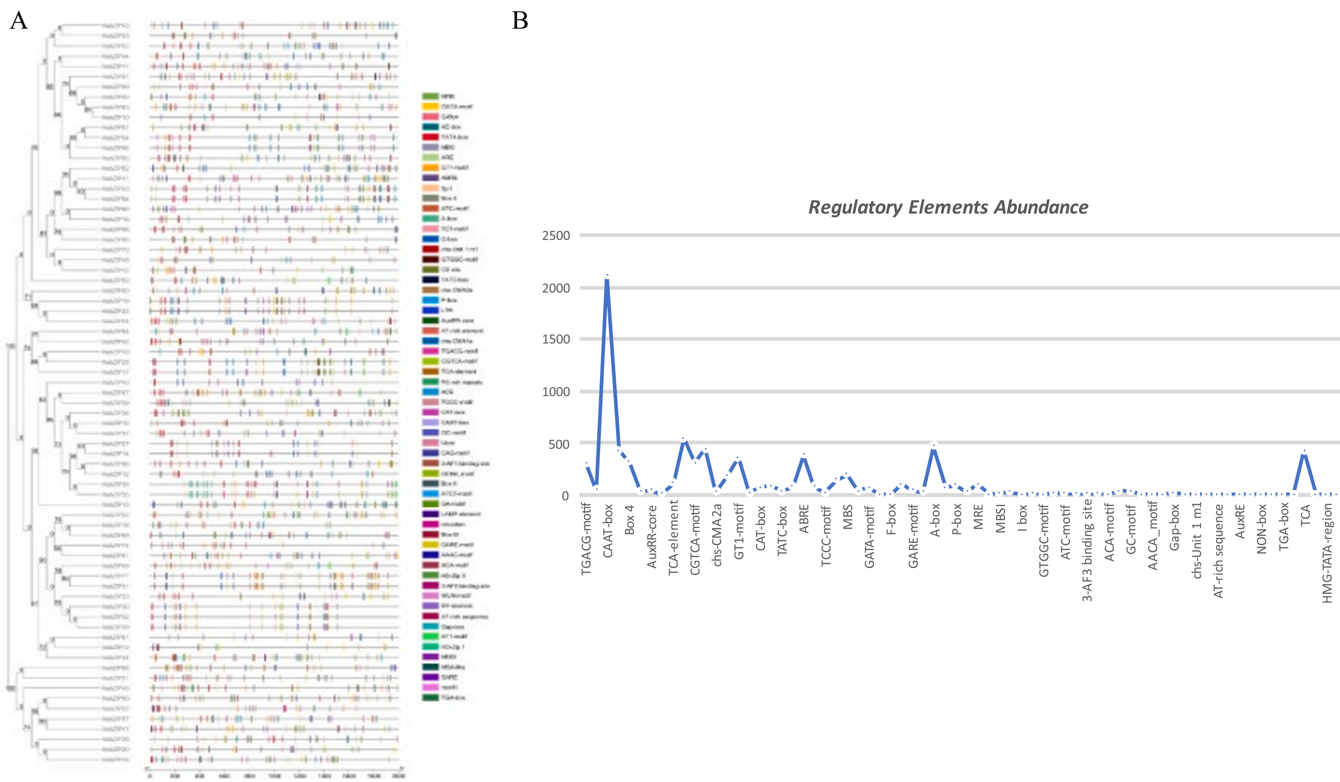

Fig. 5. (A) C is Acting Regulatory Elements of HabZIP genes are shown in figure according to phylogenetic tree the genes are organized as well and in the ligand figure colors of Cis Acting regulatory elements are shown. (B) Regulatory Elements abundance.

2023).

Phylogenetic conservation and implications for functional adaptation

Overall, 73 members of bZIP were identified in Sunflower. The members of this gene family were also identified in a diverse range of plants including *A. thaliana* (Dröge-Laser et al., 2018)*, O. sativa* (Corrêa et al., 2008)*, Z. mays* (Wei et al., 2012), *V. vinifera* (LIU et al., 2014b) and *B. distachyon* (Liu and Chu, 2015). Similarly, a variable number of bZIPs were identified in various plants that were not associated with their genome size or taxonomic group. However, the number of bZIPs is associated with the level of genome ploidy or mode of duplication (Azeem et al., 2018; Yue et al., 2023). It was also inferred previously that higher plants with more bZIP genes had stronger environment adaptation abilities (Liu and Chu, 2015).

The phylogenetic analysis of *H. annuus* has revealed that bZIPs can be divided into 11 groups based on the classification of bZIPs in Arabidopsis. In addition to *A. thaliana* and *H. annuus, C. arabica,* L. *sativa,* and *V. vinifera.,* all are involved in the categorization of orthologs (Ahmed et al., 2021). It was observed that each clade had both AtbZIP and HabZIP genes, which indicates that the evolution of bZIP genes was conservative between these plants. Similar results were reported previously for *B. distachyon* (Liu and Chu, 2015). If a phylogenetic analysis is performed on the basis of conserved motifs, domains as well as gene structure analysis, it can strengthen the evolutionary statement of the gene family (Azeem et al., 2022). Strong phylogenetic conservation of bZIPs between *A. thaliana* and *H. annuus* suggests the existence of significant functional conservation (Fig. 1). Group-A, HabZIP35 and HabZIP38 are strong paralogues. They also share orthology with AtbZIP35, 37 and 38 (ABF1, ABF3 and ABF4) that are the principal transcription factors in ABA-mediated osmotic and salt stress signaling in vegetative tissues (Choi et al., 2000; Uno et al., 2000; Fernando et al., 2018). In group X, HabZIP60 share a close orthology with AtbZIP60, which has a distinctive role in regulating plant specific endoplasmic reticulum stress response (Iwata and Koizumi, 2005). In group-I, HabZIP18 shares a close paralogous relationship with HabZIP52 and HabZIP69. These three TFs share orthology with AtbZIP18 and AtbZIP52 that are involved in heat stress response (Wiese et al., 2021). Similarly, members of group-S are important for plant response to environmental stresses. Here HabZIP04 share a close paralogous relationship with HabZIP05, 06 and 07. Meanwhile, a paralogous relationship exists for this sub-group of HabZIPs with AtbZIP04. In *A. thaliana*, stress-responsive proteins (AtbZIP04 and AtbZIP43) were suggested to control AtbHLH109 expression (Nowak and Gaj, 2016). Similarly, HabZIP01 belongs to an orthologous group of G-box bZIPs i.e., AtbZIP02, 11 and 44 (Figure). The G-box bZIP transcription factors in Arabidopsis are key players in the regulation of stress responses by binding to G-box motifs in the promoters of stress-responsive genes and modulating their expression (Ezer et al., 2017). HabZIP08, 42, 48, 68 and 70 represent an isolated group of paralogs in group-S. They share an orthologous relationship with AtbZIP70 and atbZIP75, which are involved in ABA signaling, seed development, light response, and secondary metabolism (Liu et al., 2020). In conclusion, the identified HabZIP transcription factors show strong paralogy and share orthology with key regulatory factors in ABA-mediated osmotic and salt stress signaling, plant-specific endoplasmic reticulum stress response, heat stress response, and secondary metabolism (Choi et al., 2000; Uno et al., 2000; Fernando et al., 2018; Iwata and Koizumi, 2005; Wiese et al., 2021; Nowak and Gaj, 2016; Ezer et al., 2017; Liu et al., 2020).

Protein-protein interaction analysis supported these phylogenetic relationships (Fig. 3B). Distictive clusters of proteins can be seen involving bZIPs and other proteins. In a cluster including COP1, CRY1, FHY3, HYH, CRY2, RUG1, HFR1, SPA1, ELF3, GI, BBX21, CIP8, PHYB, PHYA and HY5 (Fig. 3B), HY5 (ELONGATED HYPOCOTYL 5) a bZIP transcription factor in that plays a central role in the regulation of photomorphogenesis. It acts as a positive regulator of light signaling and is involved in the expression of light-responsive genes (Maxwell et al., 2003). Similarly, there is another cluster including ABI1, ABI2, ABI3,

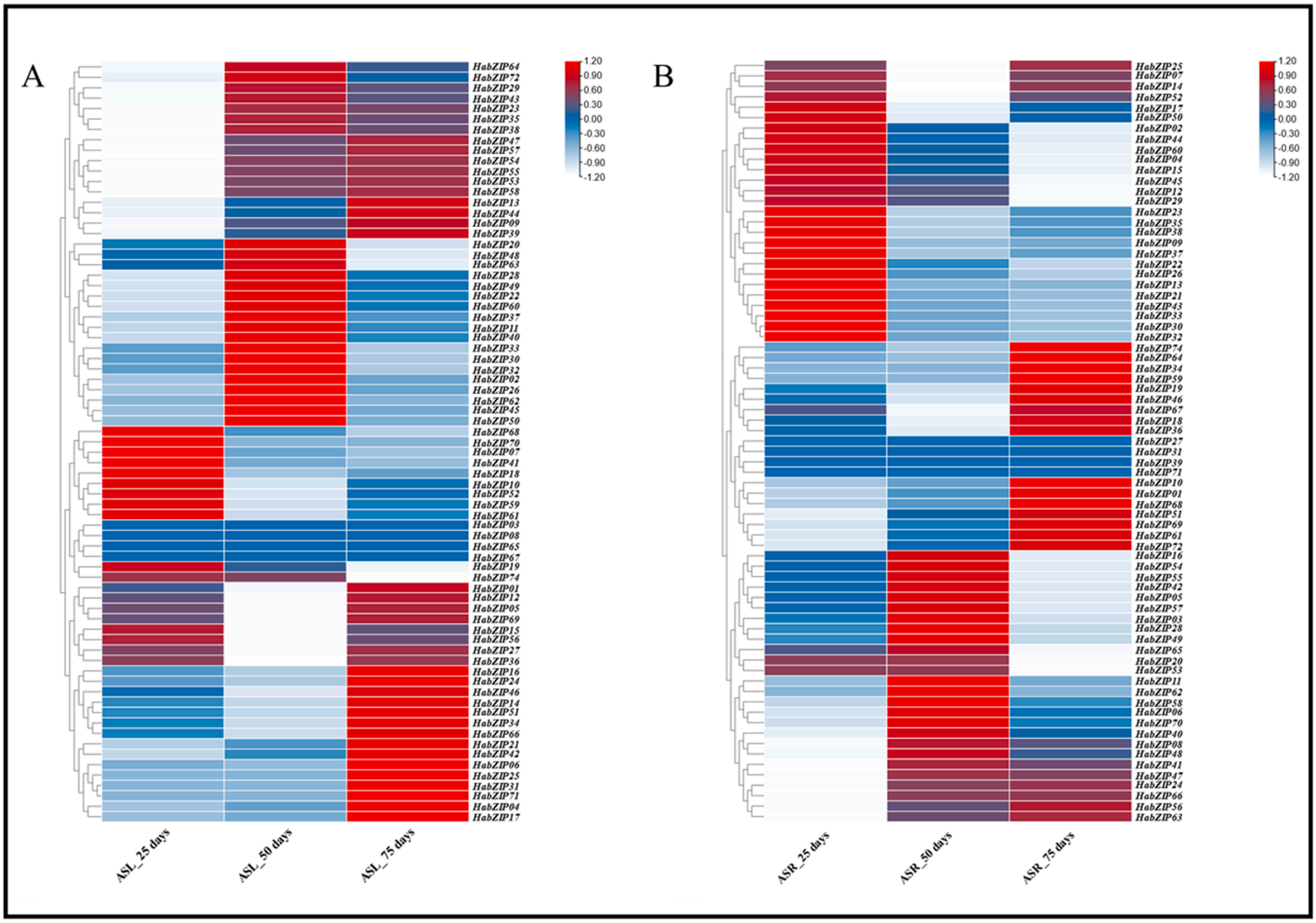

Fig. 6. (A) Expression profiles of HabZIP genes in response to Alkali stress. In the heatmap, each row represents a gene and each column represents an experiment (either control or stress). The heatmap was created by Tbtools software where log2-transformed values of relative expression levels of HabZIPs. Changes in gene expression of genes are shown in color as scale. (A) Expression profiles of HabZIPs in leaves. (B) Expression profiles of HabZIPs in roots.

ABI5, OST1, Cul4, KEG, HAB1, and PP2C. Here, AtbZIP67 (ABI5; ABSCISIC ACID-INSENSITIVE 5) acts downstream of ABI3 and plays a crucial role in ABA signaling during seed germination and seedling development (Uno et al., 2000). In protein-protein interaction studies, TGA3, a bZIP transcription factor (Han et al., 2022), has been found to interact with various other proteins, including NIMIN1, NIMIN2, NIMIN3, ROXY1, GRX48, AHBP1B, OBF5, NPR1, and NPR4, indicating its involvement in diverse regulatory pathways, particularly in plant defense responses and stress signaling (Fig. 3).

Expression analysis of bZIPs under alkali stress in H. annuus: implications for salinity stress response

The present study investigated the gene expression patterns of bZIP transcription factors using NGS data analysis in leaves and roots of *H. annuus* (sunflower) under alkali stress. The results shed light on the potential role of bZIPs in the plant's response to salinity stress.

In leaves, the bZIPs displayed a dynamic response pattern, organized in clusters during different time points of the experiment. This indicates that specific groups of bZIPs may be involved in distinct regulatory processes at different stages of alkali stress (Fig. 1). Interestingly, none of the bZIP genes were consistently upregulated throughout the entire experiment. This suggests that the regulation of bZIPs might be a tightly controlled process, with different members being activated or repressed at specific time points to modulate the plant's response to salinity stress (Weltmeier et al., 2006; Liu et al., 2014a). Additionally, the significant downregulation of four bZIPs, including HabZIP08, throughout the entire experiment suggests their potential role as negative regulators during alkali stress in leaves (Liu et al., 2012). Such negative regulators might act as crucial modulators in fine-tuning the plant's stress response to prevent excessive damage and maintain cellular homeostasis.

In roots, the expression profiles of bZIP transcription factors were categorized into early and late response clusters. Notably, around 20 bZIP genes, including HabZIP23, HabZIP30, and HabZIP60, were upregulated after 25 days of alkali stress exposure in roots. Similarly, an almost similar number of bZIPs were upregulated after 50 days and 75 days of stress exposure. This dynamic pattern of upregulation in roots suggests that these bZIPs play a vital role in the early stages of the plant's response to salinity stress and might be involved in initiating adaptive mechanisms in roots (Estes et al., 2010). The observed temporal regulation of these clusters further implies the importance of bZIPs in coordinating the plant's response to salinity stress in a stage-specific manner.

Furthermore, the real-time amplification (qRT-PCR) analysis of selected HabZIP genes in leaves under mild and severe NaCl stress provided insights into their specific responses. Differential regulation of these HabZIPs was observed during NaCl stress. HabZIP01, HabZIP04, HabZIP18, and HabZIP48 were differentially upregulated in response to both mild and severe stress conditions. On the other hand, HabZIP05, HabZIP06, and HabZIP42 showed upregulation only under severe stress. HabZIP60 exhibited an interesting pattern, being upregulated under mild stress but downregulated under severe stress. Notably, HabZIP08 was downregulated in response to both mild and severe stress conditions. These results highlight the diverse roles of bZIPs in the plant's

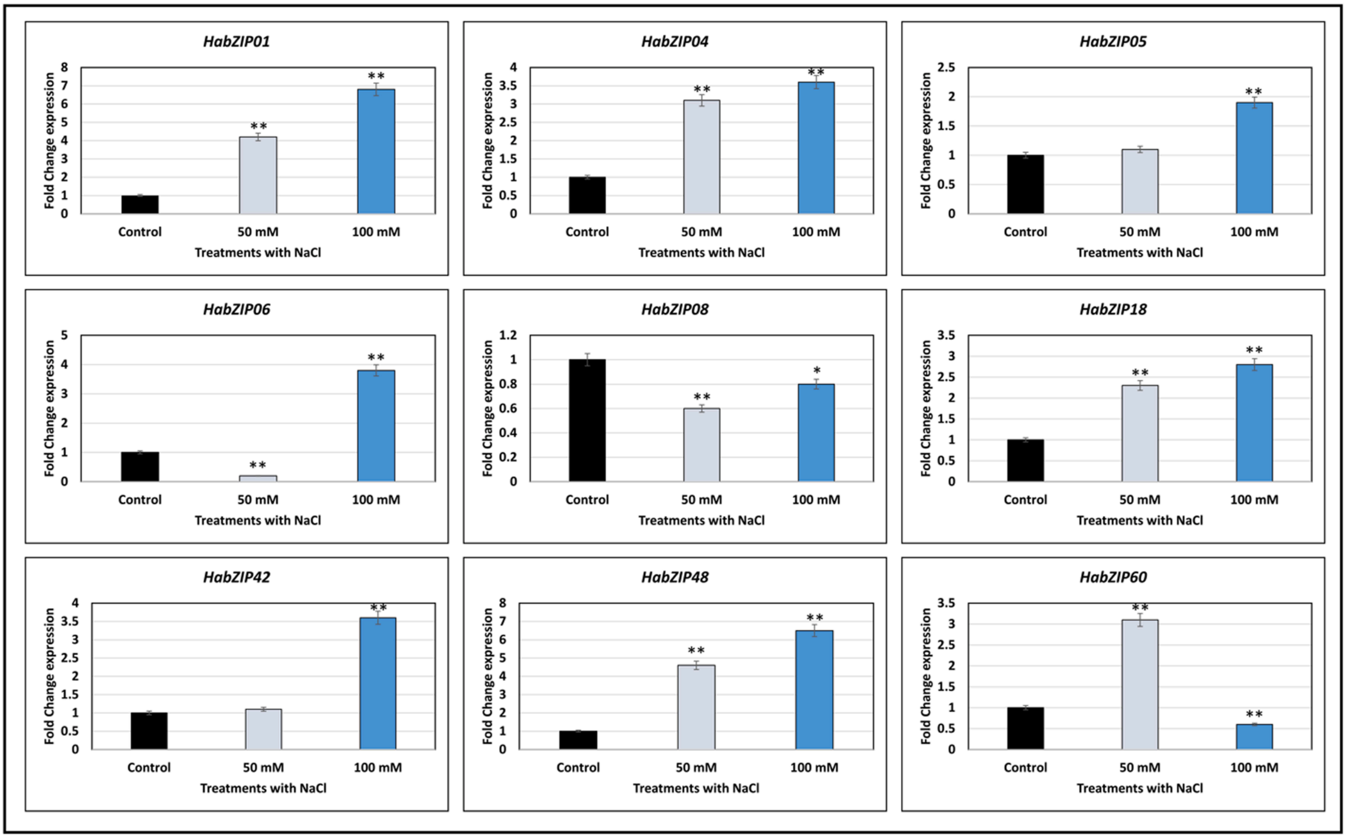

Fig. 7. RT-qPCR analysis of bZIPs in leaves of *H. annuus* in response to 50 mM and 100 mM NaCl stress. Each bar was the mean ± SE from three repeats. Asterics letters (*) and (**) in the error bar showed significant differences (Unpaired *t*-test two-tailed P value of >0.0001 and <0.0001) between the control and stressed sample groups.

response to varying degrees of salt stress (Pérez-Pérez et al., 2022). The differential expression patterns of these bZIPs suggest that they might function as specific regulators of different stress-responsive pathways, contributing to the plant's adaptation to the changing salinity conditions (Liu et al. 2023).

These results provide valuable insights into the potential molecular mechanisms underlying sunflower's response to salinity stress and pave the way for further functional characterization of the identified bZIP genes in stress tolerance and signaling pathways in sunflower. Understanding the precise roles of these bZIP transcription factors in salinity stress response may have practical implications in improving crop resilience and yield under adverse environmental conditions. Further studies investigating the functional significance of specific bZIPs and their interaction networks will be crucial for a comprehensive understanding of the complex regulatory mechanisms that govern salinity stress responses in sunflower.

Conclusion

In conclusion, this study characterized 73 bZIP genes in sunflower and identified their role in stress responses. The phylogenetic analysis grouped them into 11 clusters, while protein-protein interaction analysis revealed five distinct groups of HabZIPs. RNA-seq and qRT-PCR analysis indicated that specific bZIP clusters were involved in stress responses, with potential implications for sunflower's adaptation to abiotic stress. These findings offer valuable insights for further research and crop improvement strategies.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.stress.2023.100204.

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